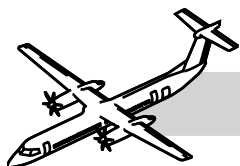


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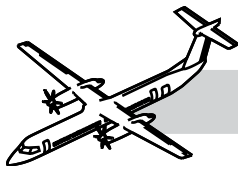
ENGINE AIR

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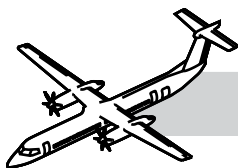


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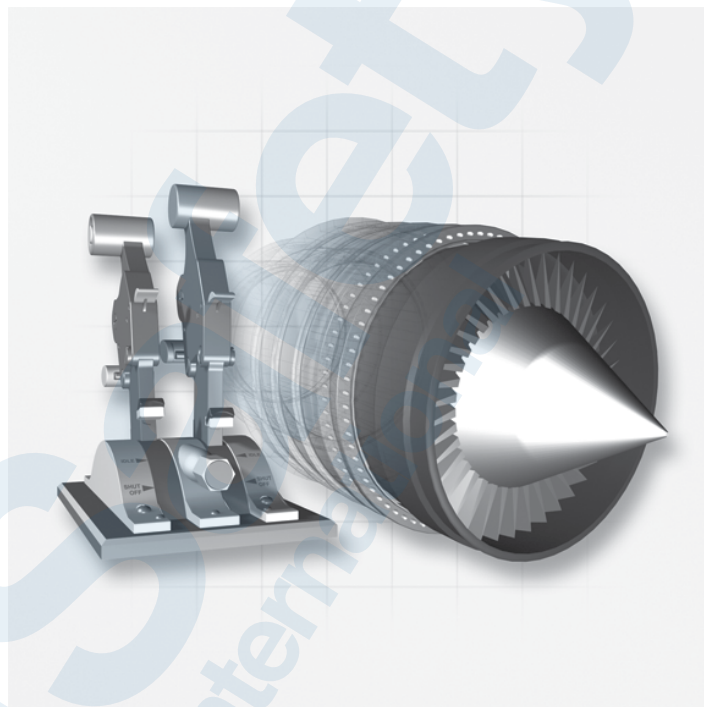


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CHAPTER 75 ENGINE AIR



75 ENGINE AIR

75-00-00 INTRODUCTION

The engine air system is designed to control the performance of the engine compressors during different flight regimes so as to prevent engine surging and stalling. It also provides compressed air to aircraft services and for sealing internal engine components.

GENERAL

Refer to Figure 75-1. Air System General.

The Air System provides compressed air for:

- Cooling of hot engine components
- Bearing sealing
- Reference air pressure for the oil pressure regulating valve
- Scavenging of various bearing cavities
- P2.2 bleed valve operation
- P2.7 bleed valve operation
- P3 signal to the FADEC.

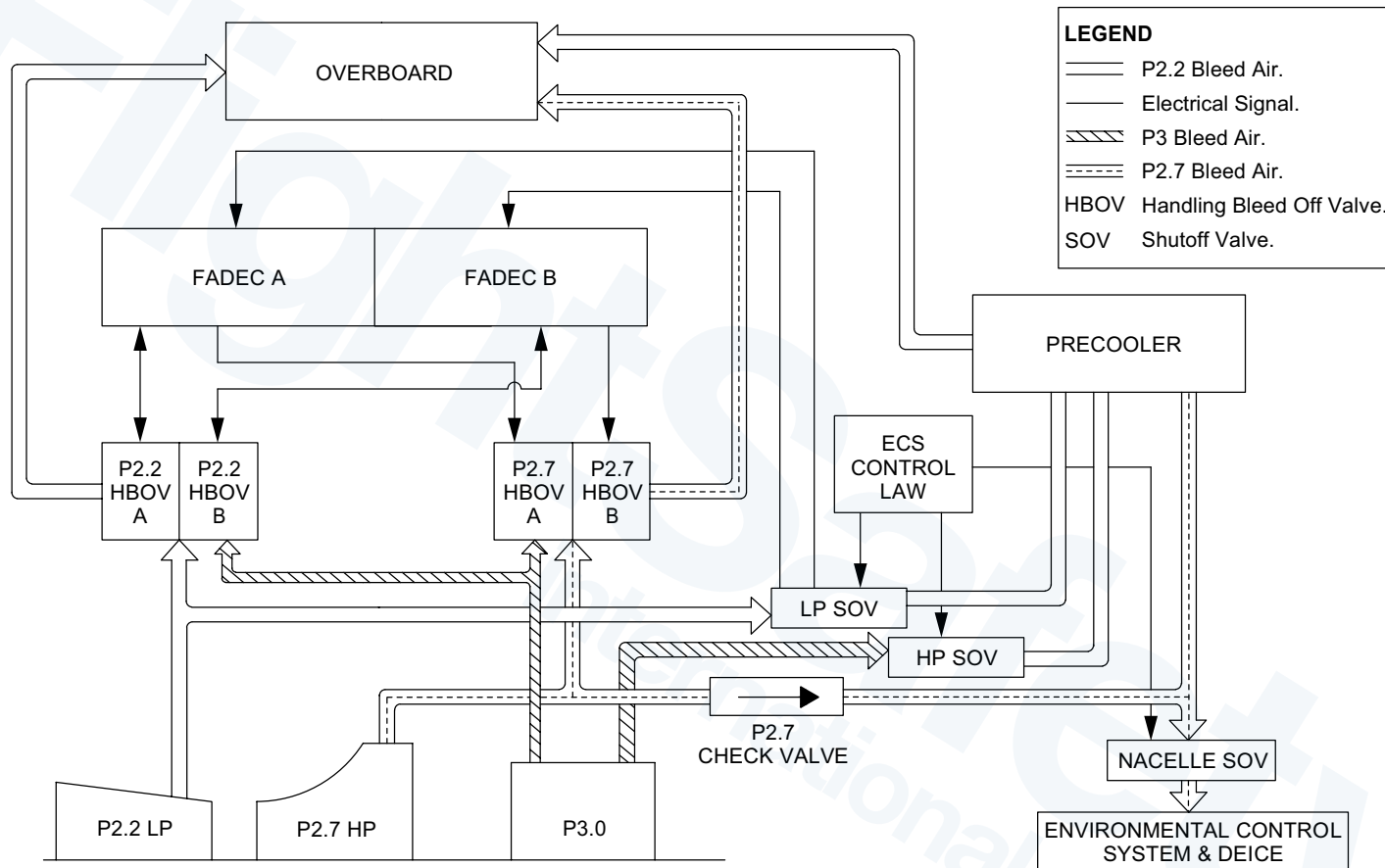
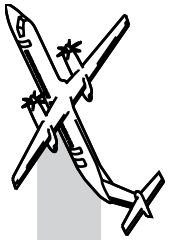
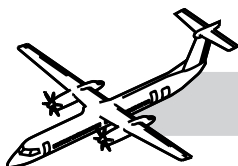


Figure 75-1. Air System General



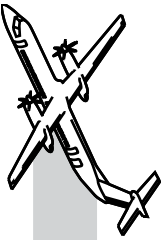
SYSTEM DESCRIPTION

NOTES

The air system is a critical system for the operation and performance of the engine.

The system comprises of these sub-systems:

- Cooling and distribution
- Compressor control
- Air indicating.



75 ENGINE AIR

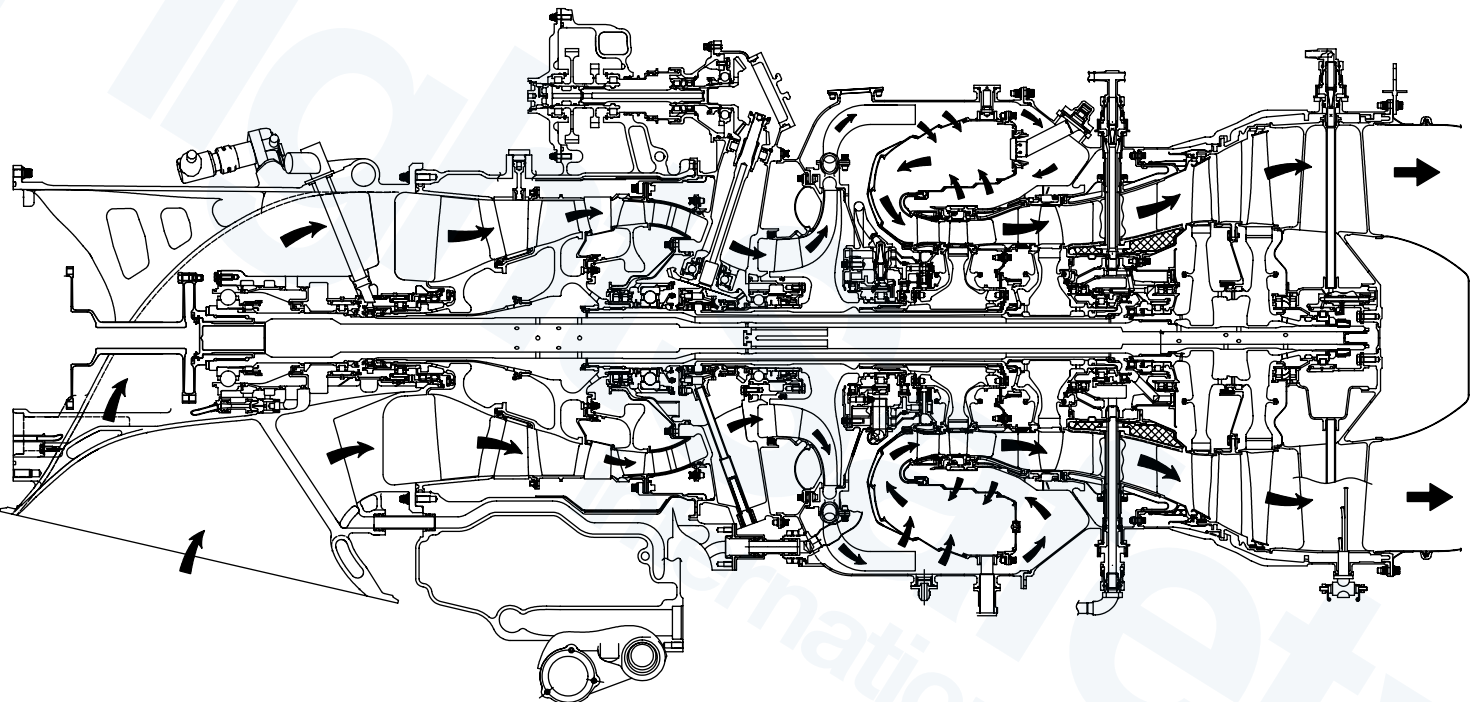
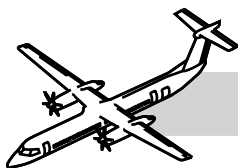


Figure 75-2. Engine Airflow



75-20-00 COOLING AND DISTRIBUTION

NOTES

GENERAL

Refer to Figure 75-2. Engine Airflow.

The cooling and distribution system is a critical system for the operation and performance of the engine. During engine operation air is drawn into the engine and compressed. Some of this air is diverted through internal passages and external tubes. This air is used to cool engine components, seal and scavenge bearing oil cavities and for aircraft services through the environmental control system. Engine cooling is a critical system for the operation and performance of the engine. The engine Cooling System is designed to control the temperature of critical engine components. It also provides air for engine sealing and aircraft services.

The function of the distribution system is performed by the:

- Internal air passages
- ICC to Turbine Support Case (TSC) cooling air tube.

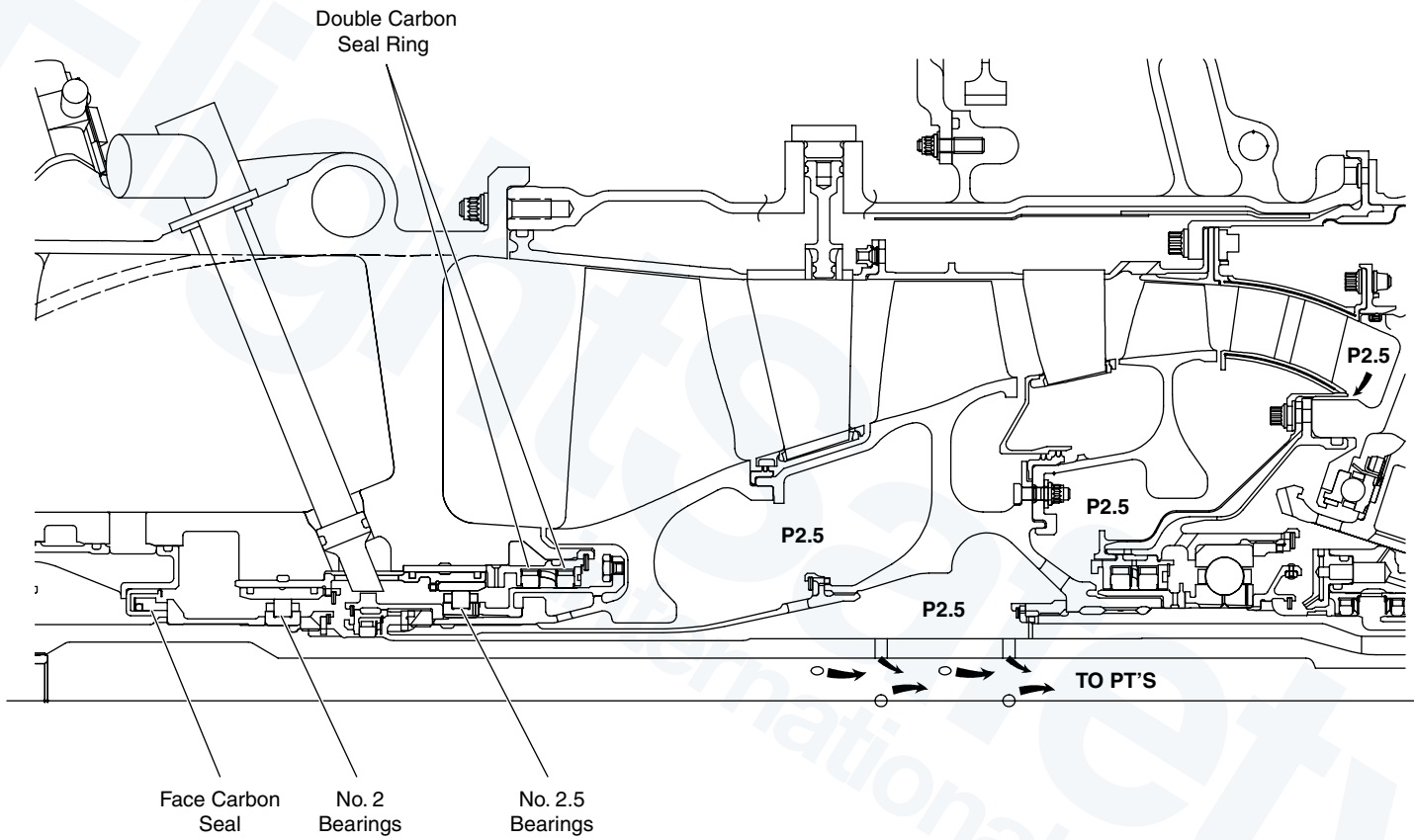
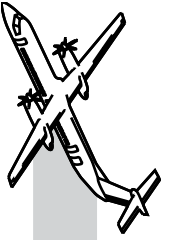
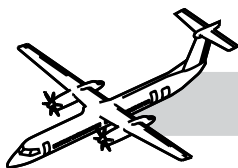


Figure 75-3. Distribution to No. 2 and 2.5 Bearing Cavity



Internal Air Passages

Refer to:

- Figure 75-3. Distribution to No. 2 and 2.5 Bearing Cavity.
- Figure 75-4. LP Turbine Blade Cooling.
- Figure 75-5. H.P Turbine Cooling.
- Figure 75-6. Vane Ring Cooling.
- Figure 75-7. Distribution to No. 3, 4 and 5 Bearing Cavity.
- Figure 75-8. Distribution to No. 6, 6.5 and 7 Bearing Cavity.

Internal air passages direct air flow to seal and scavenge the bearing and seal oil cavities. This air is also used to cool internal engine components.

The axial flow compressor is cooled internally using P2.5 air. the P2.5 air leaks into the compressor disc after the last stage of stator blades. From the compressor disc the P2.5 cooling air flows through drilled passages into the turbine shaft and then flows aft to the power turbines. The air also leaks across the carbon seals at No.2 and No. 2.5 bearings to prevent the lubricating oil from leaking away from the bearings.

The HP and LP turbines are cooled by P3 air flowing in through the blade tips. This P3 air is from the cooling airflow of the combustion chamber liner. P2.5 air flowing through passages from the drive shaft is used to seal the bearing cavities of No. 3, 4 and 5 bearings. P2.5 air is also used to cool the front face of the LP turbine and the rear face of the LP turbine.

P2.5 air flowing from the turbine shaft is used to cool the discs and the roots of the blades of the power turbines. P2.7 air is used to cool the power turbine blades internally.



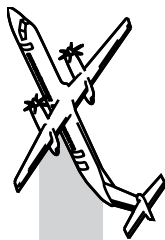
Figure 75-4. LP Turbine Blade Cooling



Figure 75-5. H.P Turbine Cooling



Figure 75-6. Vane Ring Cooling



75 ENGINE AIR

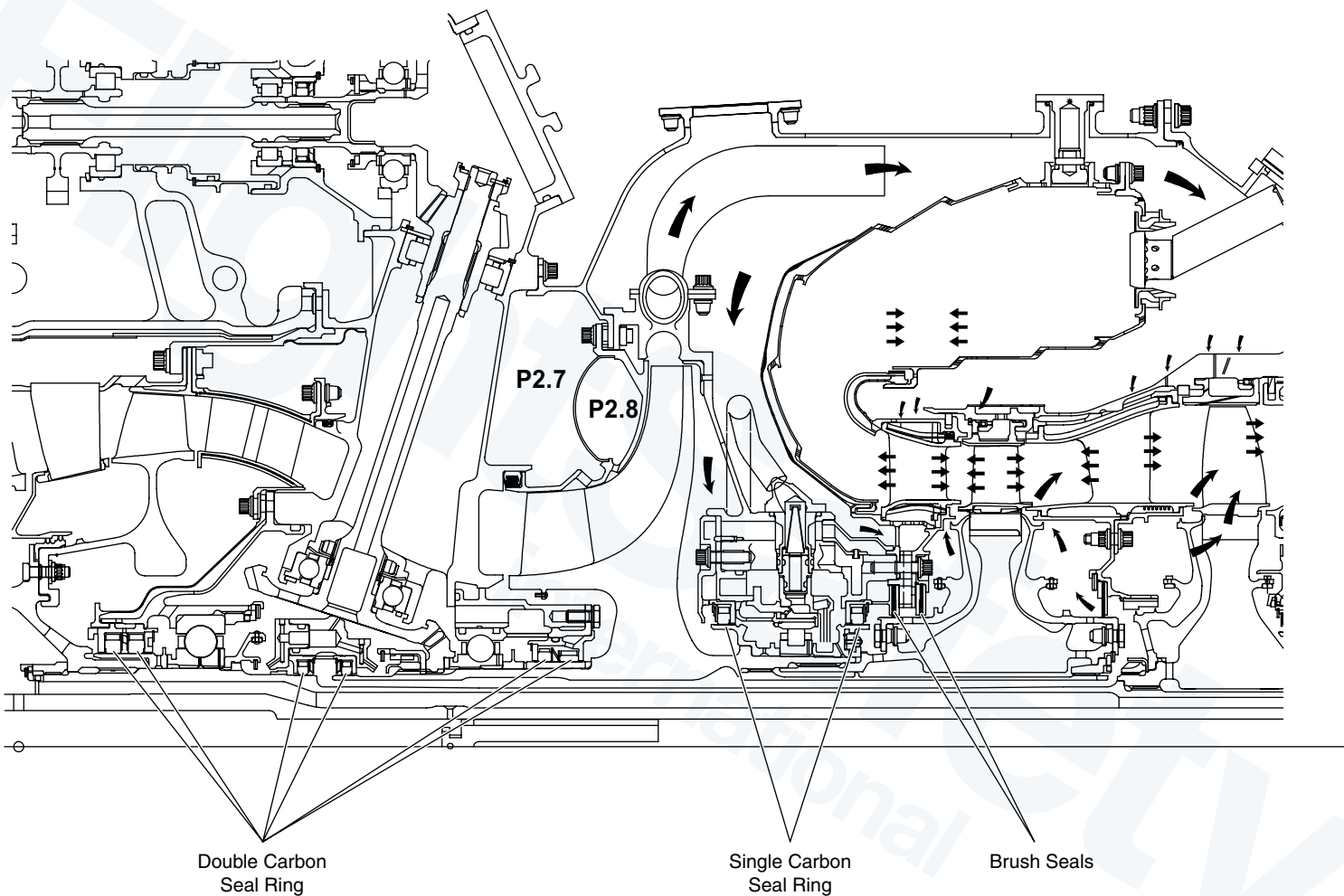


Figure 75-7. Distribution to No. 3, 4 and 5 Bearing Cavity

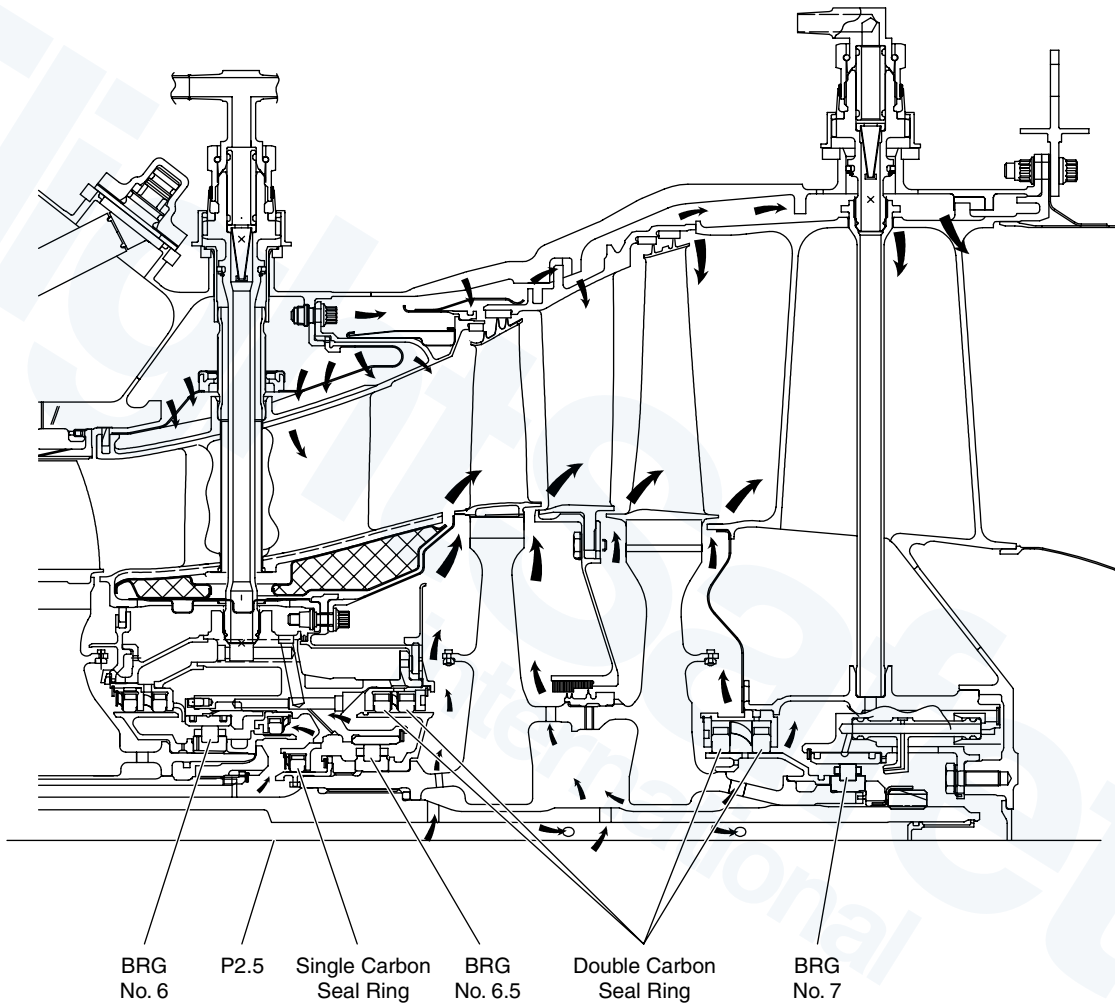
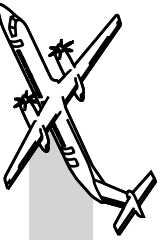


Figure 75-8. Distribution to No. 6, 6.5 and 7 Bearing Cavity

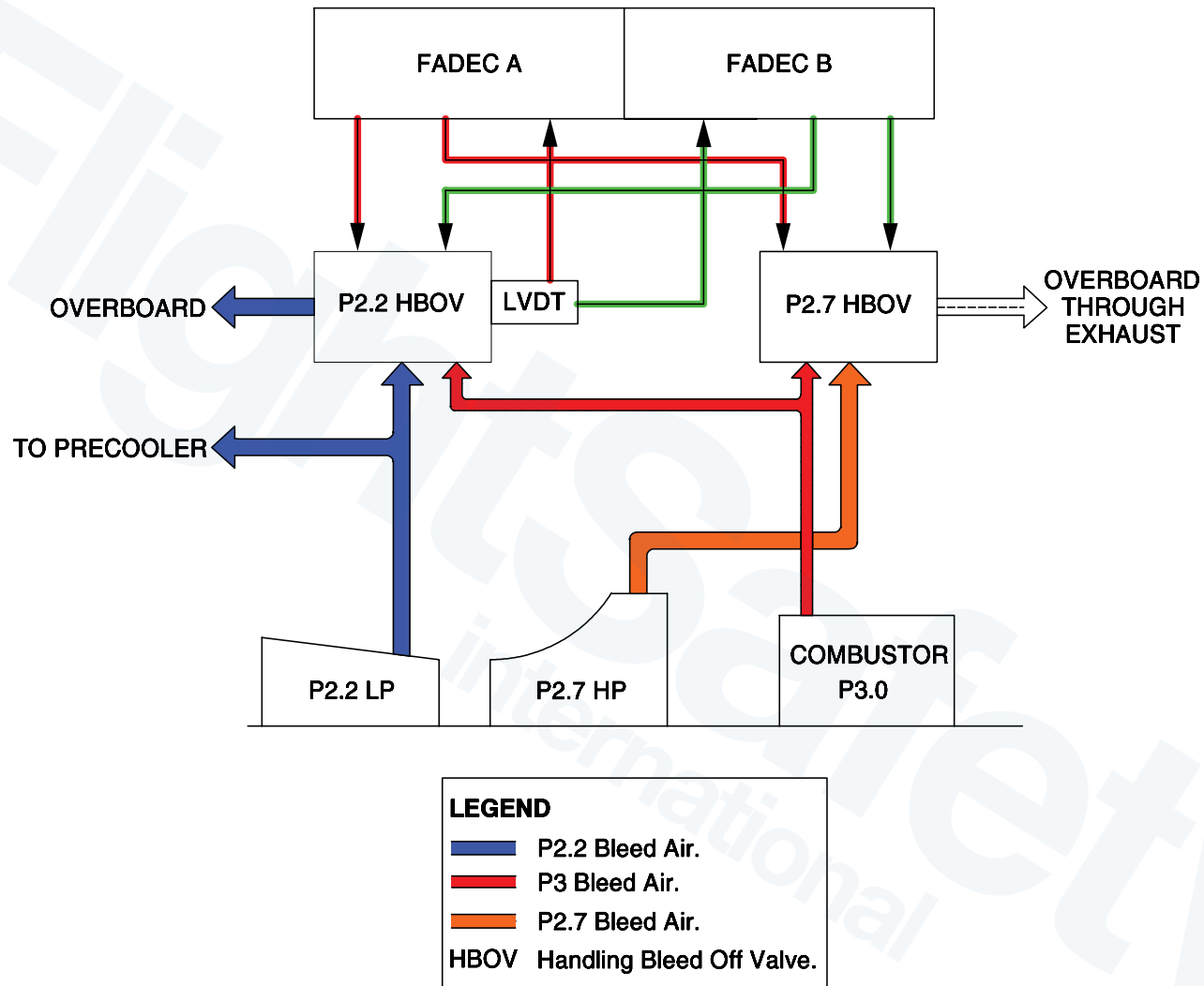
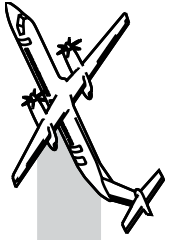
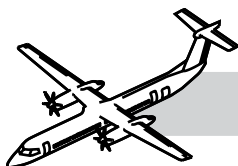


Figure 75-9. Compressor Surge Control System



75-31-00 AIR COMPRESSOR CONTROL

NOTES

GENERAL

The air compressor control system controls the performance of the compressor.

Refer to Figure 75-9. Compressor Surge Control System.

The system is comprised of the following components:

- P2.7 Handling Bleed Off Valve (HBOV)
- P2.7/P3 check valve
- P2.2 interstage bleed valve
- P3 air separator
- Air gas generator case to P3 separator air tube
- Air-centrisep to P2.2 and P2.7 valves.

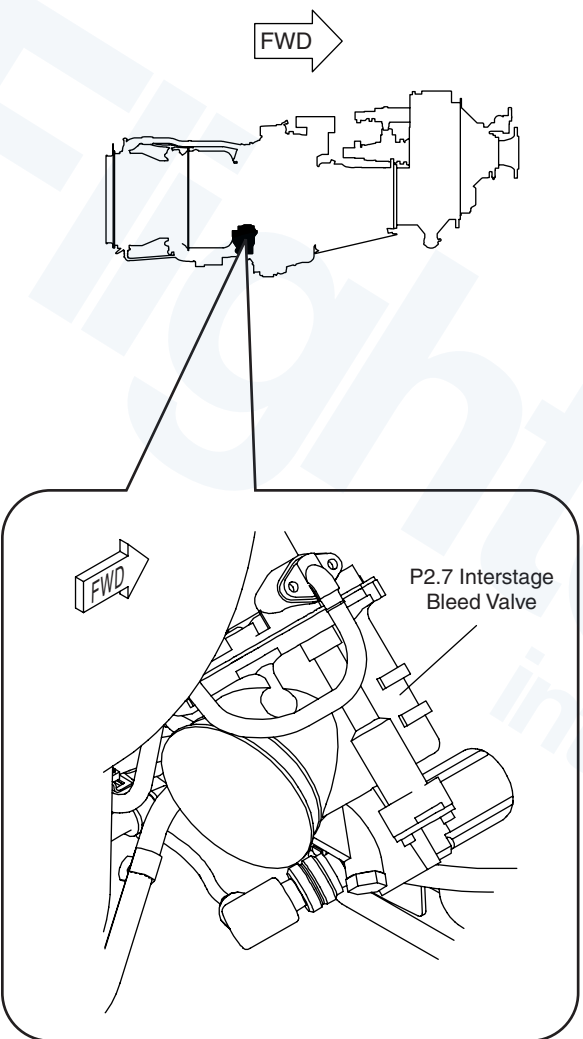
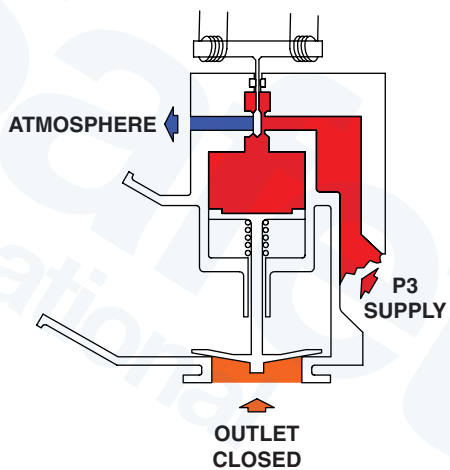
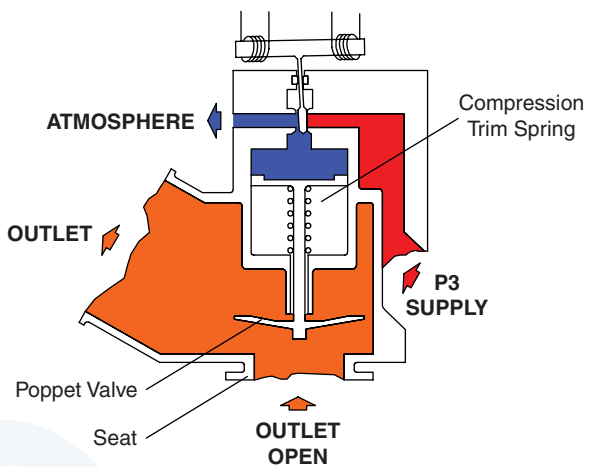
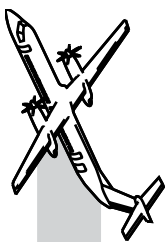
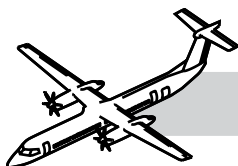


Figure 75-10. P2.7 Handling Bleed Off Valve (HBOV)



COMPONENT DESCRIPTION

NOTES

P2.7 Handling Bleed Valve (HBOV)

Refer to Figure 75-10. P2.7 Handling Bleed Off Valve (HBOV).

The P2.7 HBOV is on the right side of the ICC.

It prevents compressor stall and surge and is a normally open, dual coil torque motor driven, on-off in-line valve.

It has an:

- Inlet
- Outlet
- Flange mounted servo port.

An electrical connector is hard mounted to the torque motor.

FUNCTIONAL CHECK OF P2.7 HBOV

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

1. Open and tag the CBs in the sequence shown in task sheet
2. Do a visual inspection of the valve
3. Do an electrical resistance check of the valve with control harness connected
4. Do an electrical insulation check of the valve with control harness connected
5. Close the CBs in the sequence shown in the task sheet.

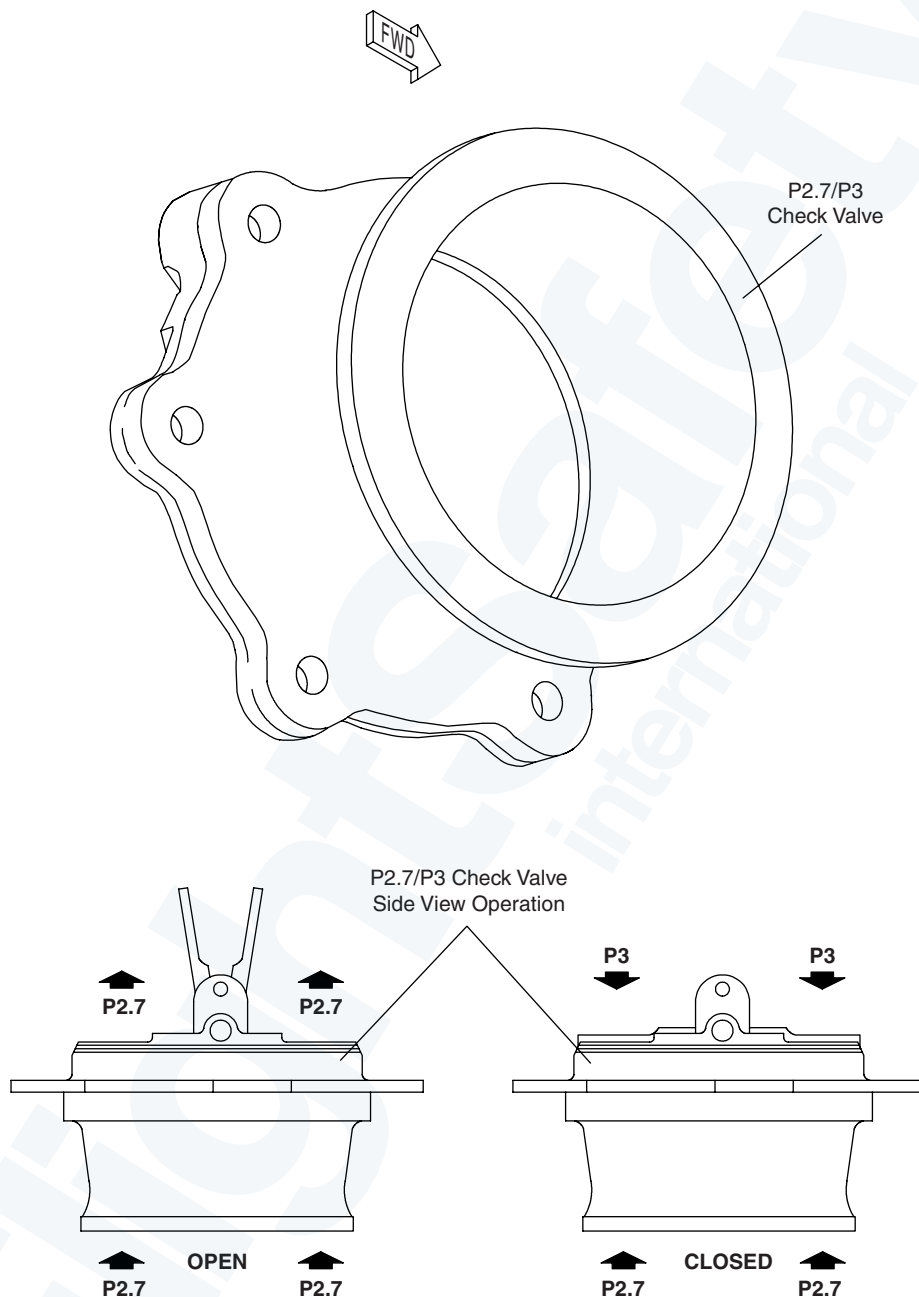
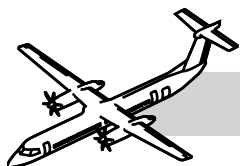
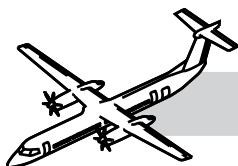


Figure 75-11. P2.7/P3 Check Valve



P2.7/P3 Check Valve

NOTES

Refer to Figure 75-11. P2.7/P3 Check Valve.

The P2.7/P3 Check Valve is a butterfly type valve installed on the ICC of the engine.

The P2.7/P3 check valve will prevent backflow into the high pressure compressor when P3 air is being used.

The P2.7/P3 check valve is part of the ECS. The ECS uses either P2.7 or P3 air as supply.

At higher engine power settings, the ECU closes the HPSOV to select the low stage (P2.7) bleed air. With the HPSOV closed, the P2.7 bleed air pressure opens the P2.7 check valve. This allows P2.7 bleed air to flow directly through to the ECS.

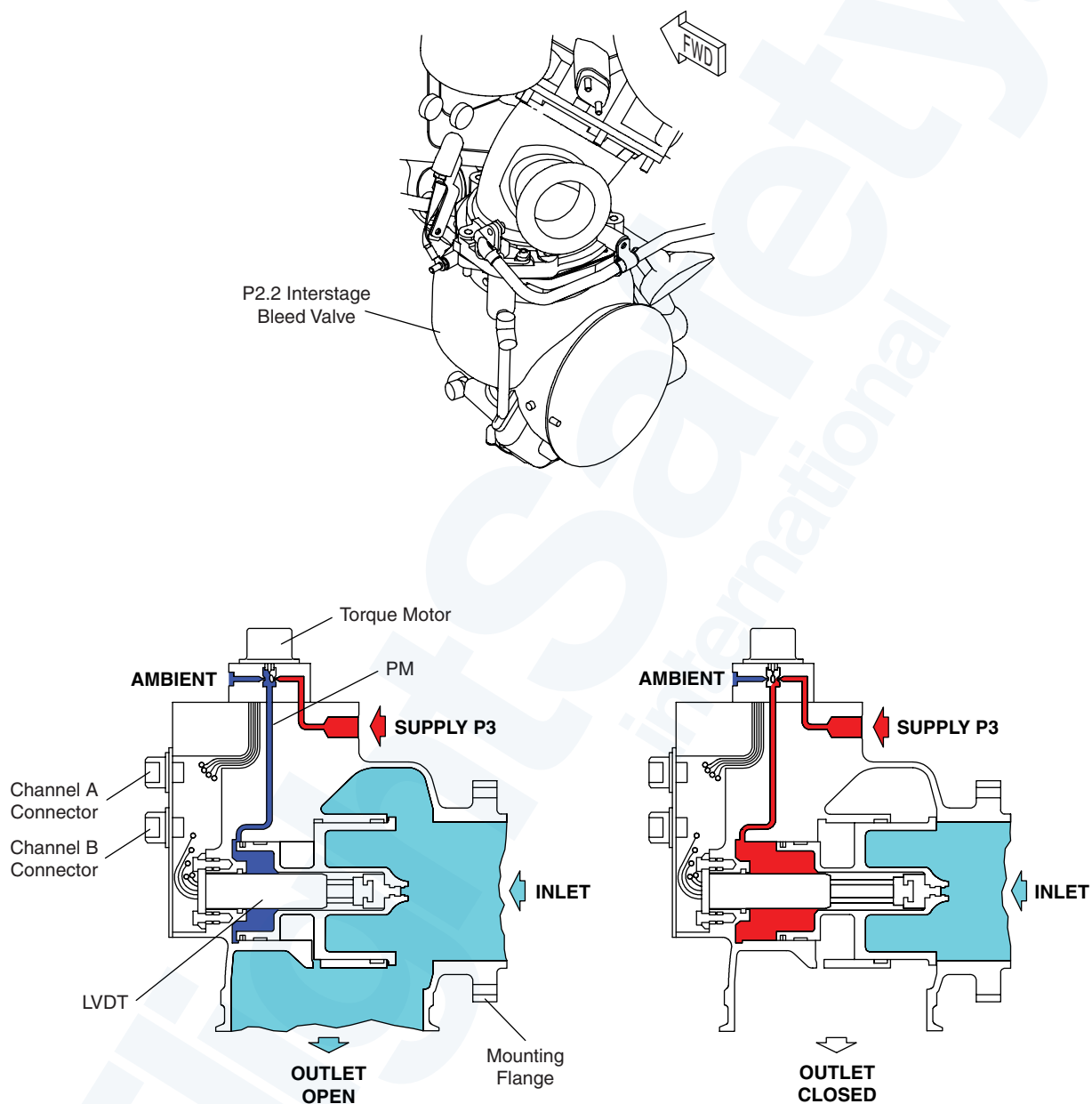
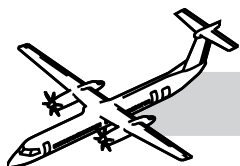
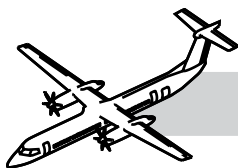


Figure 75-12. P2.2 Interstage Bleed Off Valve



P2.2 Interstage Bleed Off Valve

Refer to:

- Figure 75-12. P2.2 Interstage Bleed Off Valve.
- Figure 75-13. P2.2 Handling Bleed Off Valve Steady State Schedule.

The P2.2 valve is on the left side of the low pressure compressor case.

The P2.2 Interstage Bleed Off Valve is a pneumatically operated, dual coil torque motor actuated poppet valve. It has an inlet, outlet, flange mounted servo port and a Linear Variable Differential Transducer (LVDT) connected to the valve poppet. There is an electrical connector hard mounted to the torque motor.

The valve has a wire mesh filter upstream of the torque motor to protect the servo from contamination. The P2.2 shutoff valve is installed in line with the P2.2 low pressure ducting and the P2.2 inlet at the bottom of the precooler.

The purpose of the valve is to prevent compressor stall and surge. The valve is a normally open, modulating in-line valve.

During engine start the valve is positioned to maximum bleed. During normal operation the valve is modulated to the closed position according to a FADEC schedule with reference to N_L . The valve is positioned by a command signal from the FADEC to the valve's torque motor. This signal moves the flapper rod, which changes the control pressure to the valve, upsetting the force balance and moving the valve. The FADEC receives the LVDT position feedback signal which correlates to its schedule.

During engine shutdown the valve is commanded back to the full open position.

The P2.2 valve is designed to provide redundancy in the event of a failure. The torque motor and LVDT each have 2 coils that are controlled through 2 separately wired channels. In the event of a power loss to the torque motor, its null bias will cause the valve to open to its full bleed

position. The valve is also designed to fully open if there is a loss of servo supply pressure.

The P2.2 HBOV is fully modulating. It is controlled by the FADEC to provide surge margin during steady state operation.

The amount of valve opening is biased closed by 50% whenever the ECS is demanding P2.2 bleed for precooling P3.

This is done by transmitting a discrete signal from the ECS ECU to the ECIU and then to the FADEC on an ARINC 429 data bus.

For a transient surge the amount of opening is biased open by 50% for the duration of the surge.

For a steady state surge the amount of opening is biased open by 14%.

The FADEC will retain this bias until a FADEC Fault Clear is done. If further surging is detected the FADEC will bias the HBOV open by an additional 14% to a maximum of 70%.

BLEED OFF VALVE SCREEN CLEANLINESS CHECK

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

1. Remove the screen from the P2.2 and/or the P2.7 HBOV.
2. Place screen in the Airflow Fixture Tool.
3. Supply air through the screen and determine from the gauge the delta pressure across the screen.
4. Compare this delta pressure with the tables in the task sheet.
5. The value of the delta pressure determines the amount of contamination of the screen.
6. If necessary clean the screen either electrosonically or ultrasonically in accordance with Task 75-31-33-100-801.

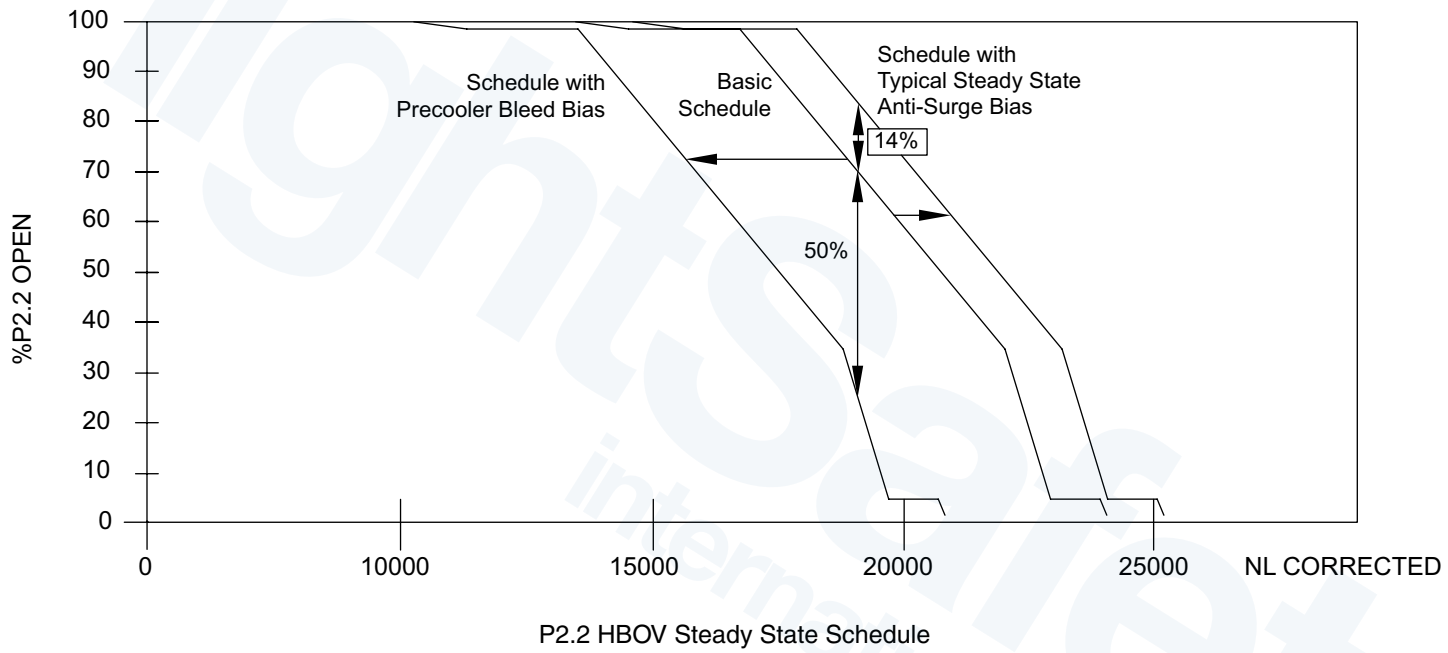
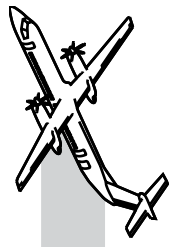
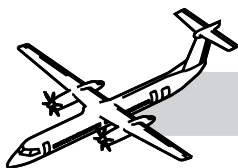


Figure 75-13. P2.2 Handling Bleed Off Valve Steady State Schedule



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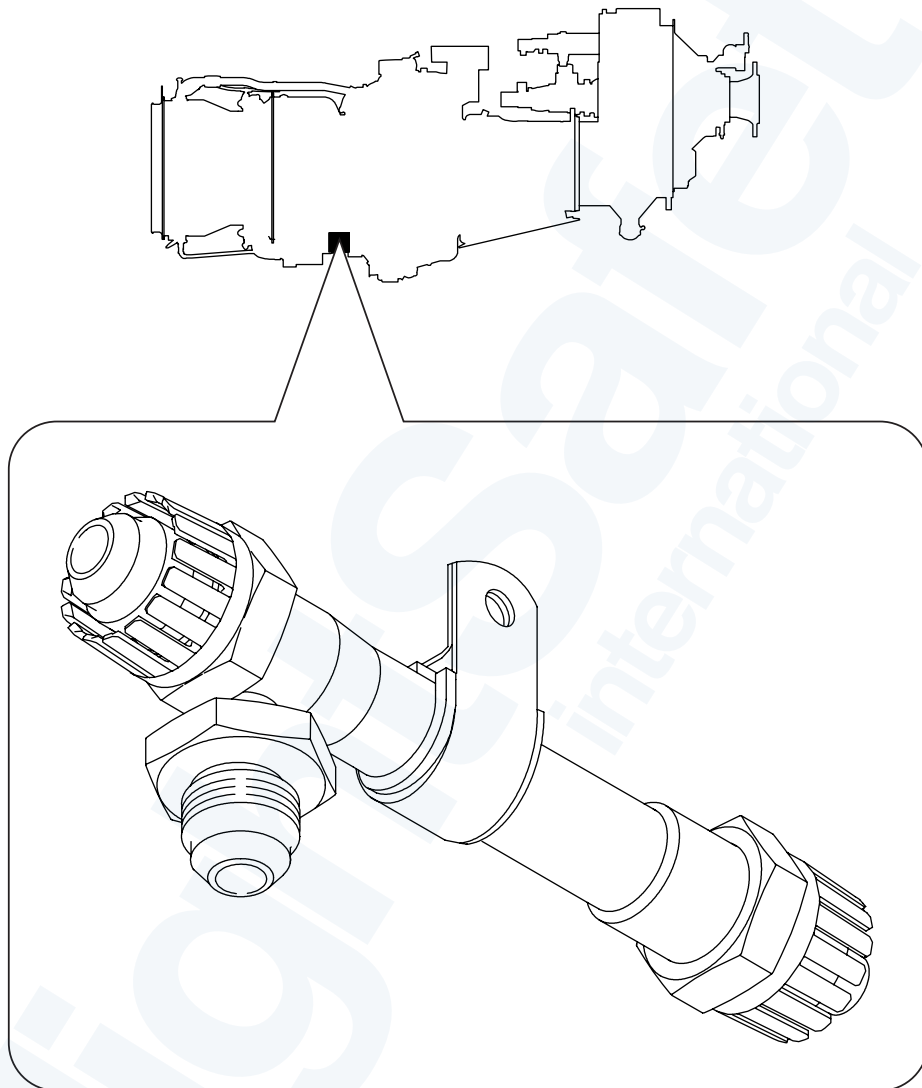
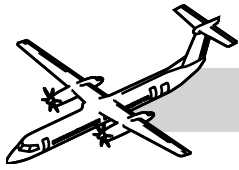
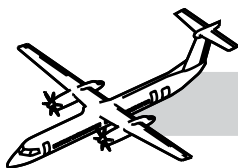


Figure 75-14. P3 Air Separator



P3 Air Separator

NOTES

Refer to Figure 75-14. P3 Air Separator.

The P3 air separator, also referred to as the centricep, is installed between the low pressure compressor case and the gas generator case.

The purpose of the separator is to prevent contamination damage of the P3 servo inlet port. A centrifugal separator is used to remove dirt and dust from the P3 air.

If the separator output fails, due to blockage, both the P2.2 and the P2.7 HBOV's will fail to the OPEN position.

FADEC will increase fuel flow, with a subsequent increase in ITT, N_H , and N_L , to match the existing power request.

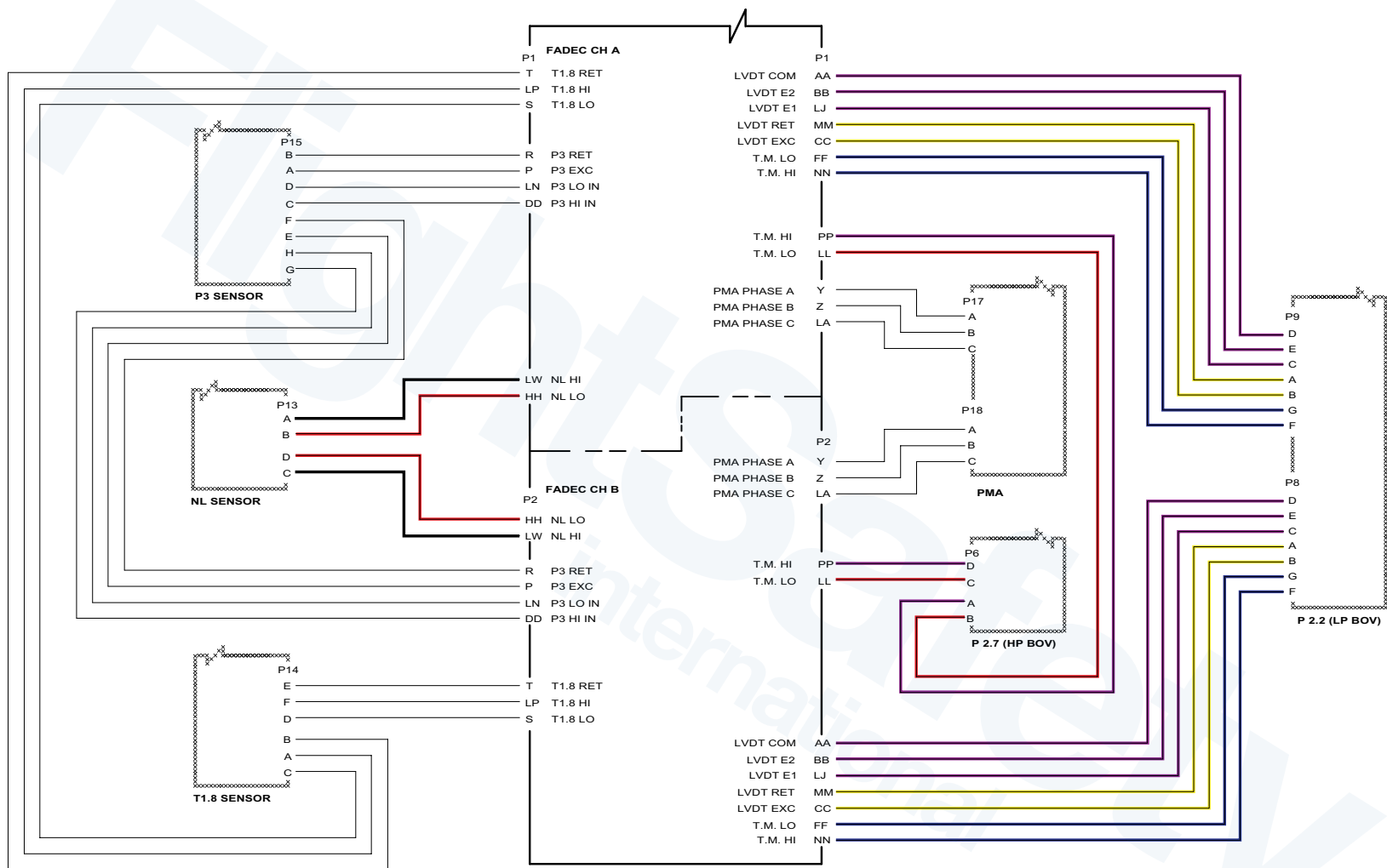
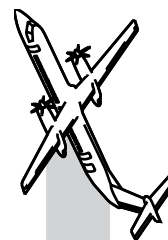
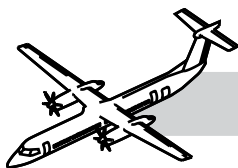


Figure 75-15. HBOV's Controls



OPERATION

Refer to Figure 75-15. HBOV's Controls.

During engine START and operation at low N_L the P2.2 handling bleed off valve is held OPEN by command from FADEC EEC.

The HBOV torque motor and LVDT receive signals through the following pins:

- Torque motor signals are from FADEC to the HBOV
- LVDT signals are to and from FADEC EEC to the HBOV.

The signals from the N_L sensor go to:

- FADEC EEC Channel A.

The signals from the N_L sensor go to:

- FADEC EEC Channel B.

HBOV Basic Schedule

As the engine accelerates the signal from the N_L sensor will increase. When the signal equals approximately 70% corrected N_L , FADEC EEC will begin to modulate P2.2 HBOV in a closing direction, by controlling the torque motor of the valve.

Pre-cooler Bleed Bias Schedule

If the P2.2 SOV in the bleed air system is OPEN, a signal from a switch on the valve will cause FADEC EEC to modulate the P2.2 HBOV closed by 50% of its normal opening schedule.

P2.7 HBOV

After engine start, with no compressor stall/surge detected, FADEC EEC will signal the P2.7 HBOV to close.

The HBOV will remain closed through the remainder of engine operation, unless FADEC EEC detects that the P2.2 HBOV is not preventing a compressor stall/surge. In this situation, FADEC EEC will command the P2.7 valve to OPEN.

Compressor Stall/Surge Detected

If compressor stall surge is detected, FADEC EEC will open the HBOV's, decrease the fuel flow and energize both spark ignitors until surge recovery is achieved.

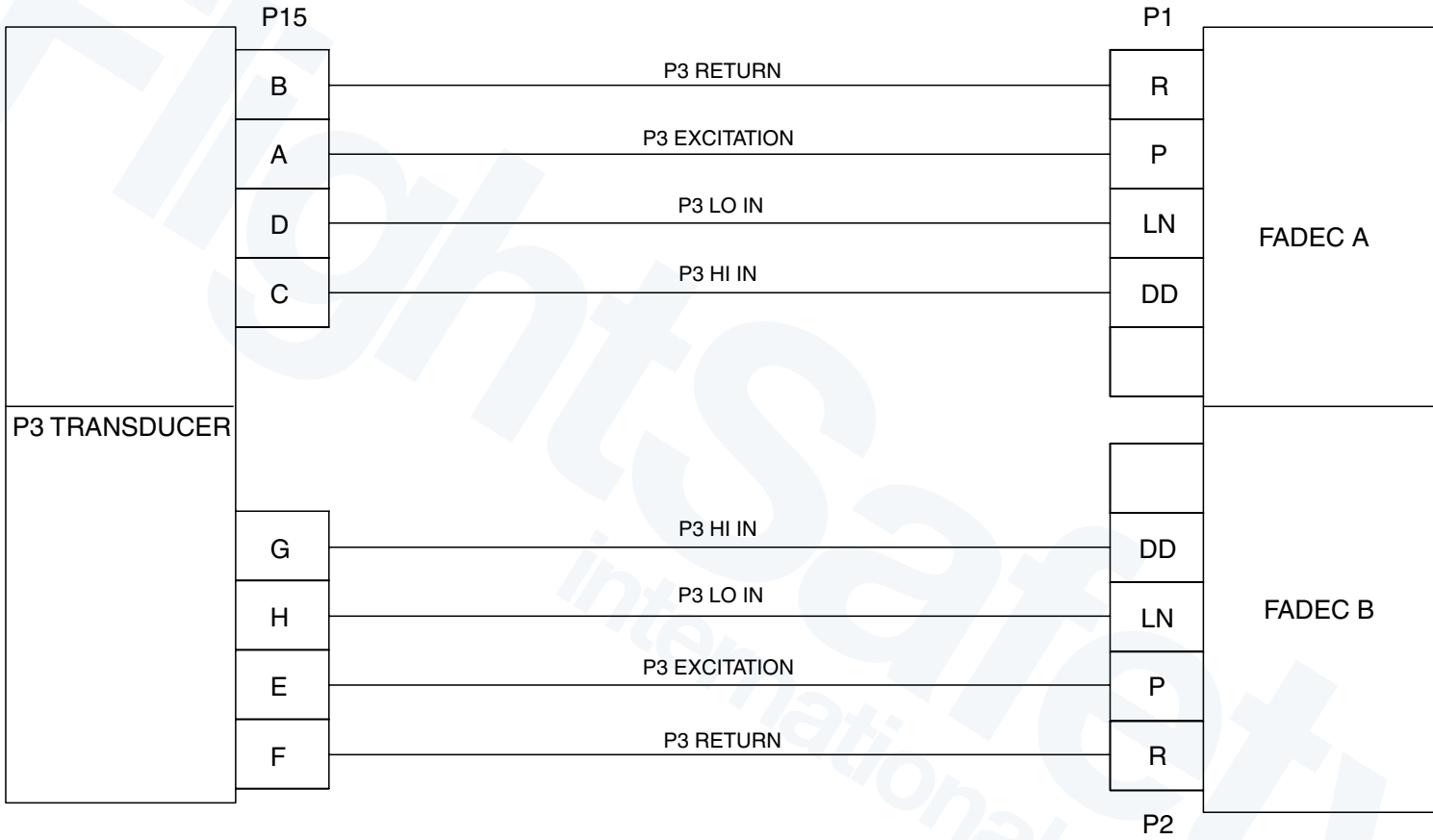
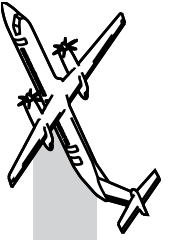
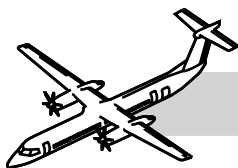


Figure 75-16. Air Indication System Schematic



75-41-00 AIR INDICATION

NOTES

INTRODUCTION

The air indication system senses compressor discharge pressure at the exit from the HP compressor and transmits this pressure, as an electrical signal, to the related FADEC.

GENERAL

FADEC uses a signal from the P3 pressure transducer for:

- Engine N_L decouple logic
- Bleed air control of the HBOV
- Surge recovery and for
- ECS HPSOV control.

A transducer senses P3 air pressure and converts this pressure to a proportional electrical signal that is then sent to the FADEC.

SYSTEM DESCRIPTION

Refer to Figure 75-16. Air Indication System Schematic.

The FADEC uses this signal for the N_L decouple logic and surge recovery.

The FADEC also transmits it to the ECS ECU, for bleed air control.

The function of the Air Indication system is performed by the:

- P3 pressure transducer and
- P3 transducer to gas generator case air tube.

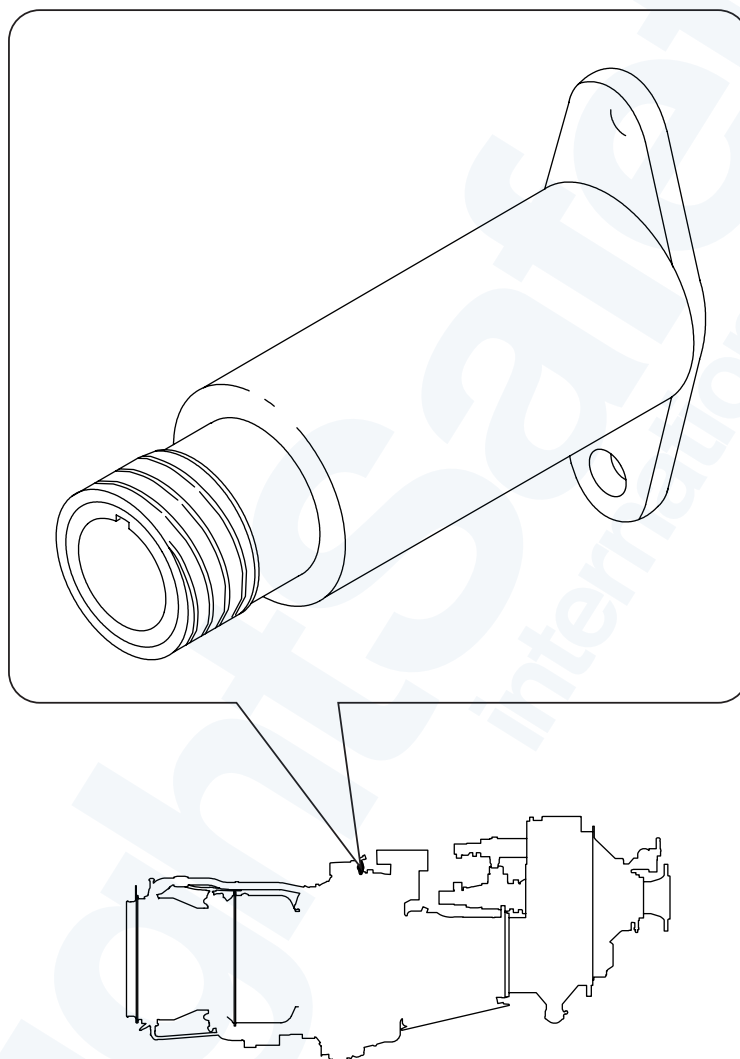
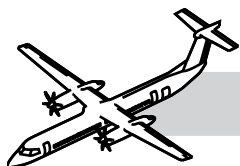
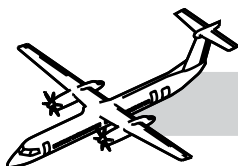


Figure 75-17. P3 Pressure Transducer



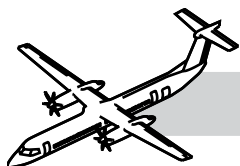
COMPONENT DESCRIPTION

P3 Pressure Transducer

Refer to Figure 75-17. P3 Pressure Transducer.

The P3 pressure transducer senses compressor discharge pressure at the exit from the HP compressor.

It is connected to the gas generator case through a stainless steel tube and is installed on the cyclonic de-aerator.



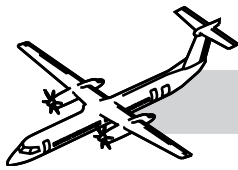
75-00-00 SPECIAL TOOL & TEST EQUIPMENT

- Barfield TT- 1000A Ohmmeter/Insulation Tester
- Simpson (or equivalent) Ohmmeter
- Commercially Available Ultrasonic or Electrosonic Bath
- PWC30128-5 Puller
- PWC59104 Air Flow Fixture - Screen Cartridge

75-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- FIM 75-00-00-810-801: Engine Performance, Increased ITT and Fuel Flow - Fault Isolation.
- AMM 75-31-33-000-801: Removal of the Intercompressor P2.2 Bleed-Off Valve Screen.
- AMM 75-31-33-400-801: Installation of the Intercompressor P2.2 Bleed-Off Valve Screen.
- AMM 75-31-33-100-801: Cleaning of the Intercompressor P2.2 Bleed-Off Valve Screen
- AMM 75-31-35-100-802: Cleaning of the P2.7 Handling Bleed-Off Valve Filter (MRB#750000-203).
- AMM 75-31-35-100-802: Cleaning of the P2.2 Handling Bleed-Off Valve Filter (MRB#750000-204).



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